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for Physics of Nanostructures

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INTRODUCTION

- Analyze mathematical models of SM in synchronous reference frames (d-q model).
- Understand the effect of cross-coupling between current components.
- Design and tune cascaded (nested-loop) control systems for SM drives.
- Implement field-oriented (vector) control with current, speed, and position loops.
- Compare dynamic responses of both control strategies and draw practical conclusions.

1 STUDY OF SM MATHEMATICAL MODEL IN THE SYNCHRONOUS REFERENCE FRAME

1.1 Task 1

1.1.1 Simulink Models

Study the influence of the current component i_{ds}^r of the current on the mechanical speed-torque characteristics of the SM.

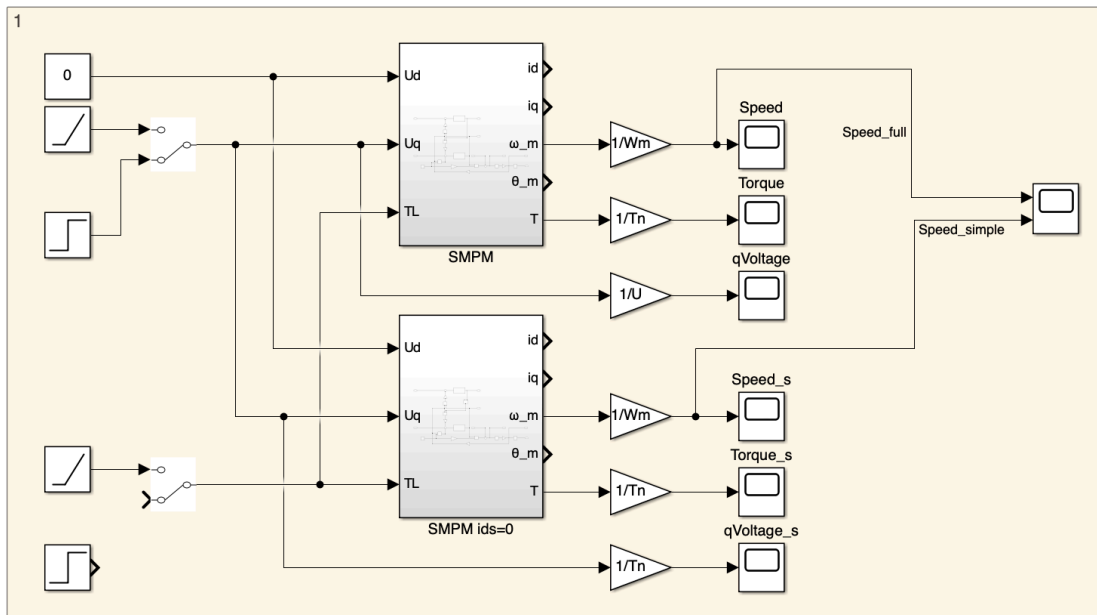


Figure 1 — Full and Simplified nonlinear d-q models

1.1.2 Simulation Results

Obtain response of $\omega(t)$ to a step reference voltage signal settling zero initial conditions $\omega_0 = 0$ and $T_L = 0$. Then after achievement steady-state mode apply step disturbance T_L

Study the influence of the current component i_{ds}^r of the current on the mechanical speed-torque characteristics of the SM.

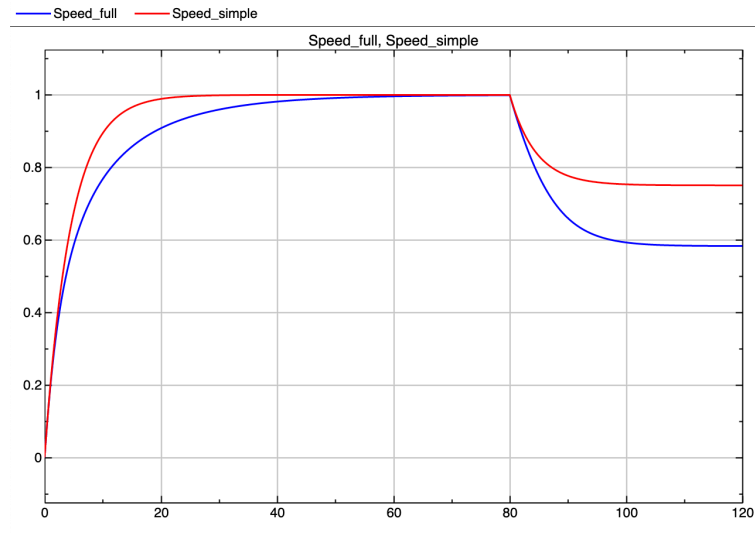
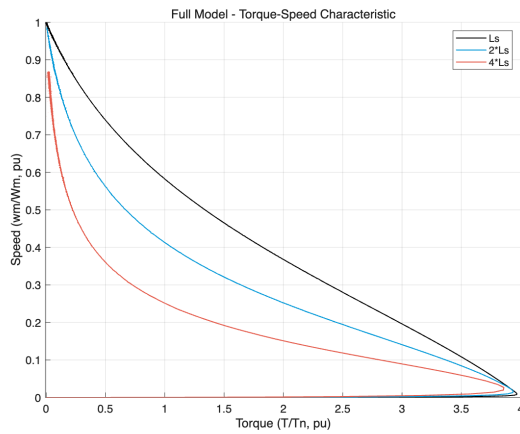
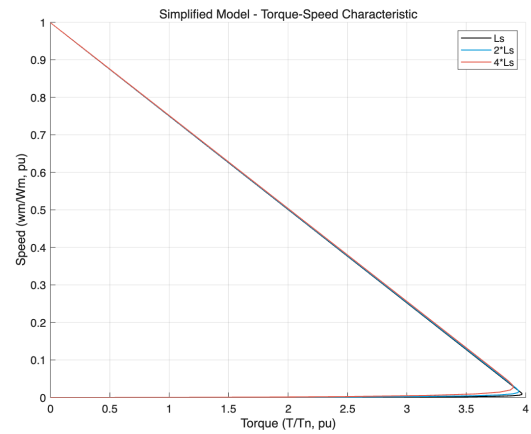


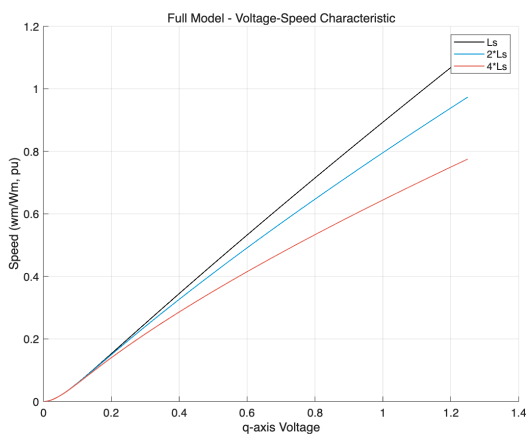
Figure 2 — Step Response of this two models



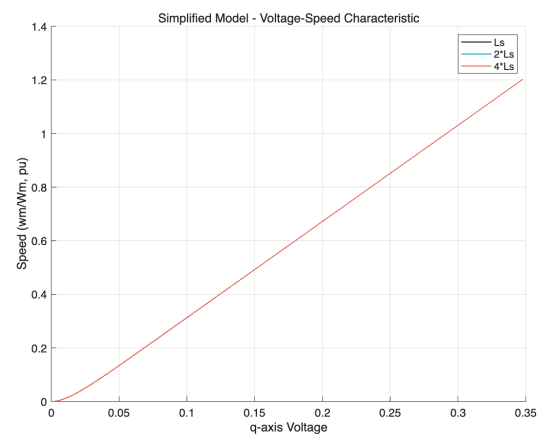
(a)



(b)



(c)



(d)

Figure 3 — Comparison of mechanical speed-torque and speed-voltage characteristics obtained from (a, c) full nonlinear d–q model and (b, d) simplified nonlinear d–q model

1.1.3 Conclusion

In the steady-state operation of the synchronous machine, the presence of the (i_d) current component leads to noticeable nonlinearities in both the speed–torque and speed–voltage characteristics. This effect becomes more pronounced as the winding inductance increases, further amplifying the influence of (i_d). To eliminate these nonlinearities and achieve linear machine behavior, it is necessary to implement a feedback control loop that maintains ($i_d = 0$).”

1.2 Task 2 : Cascaded (Nested-Loop) Control System Design

1.2.1 Torque Loop

Torque Brockett Polynomial (BP) method.

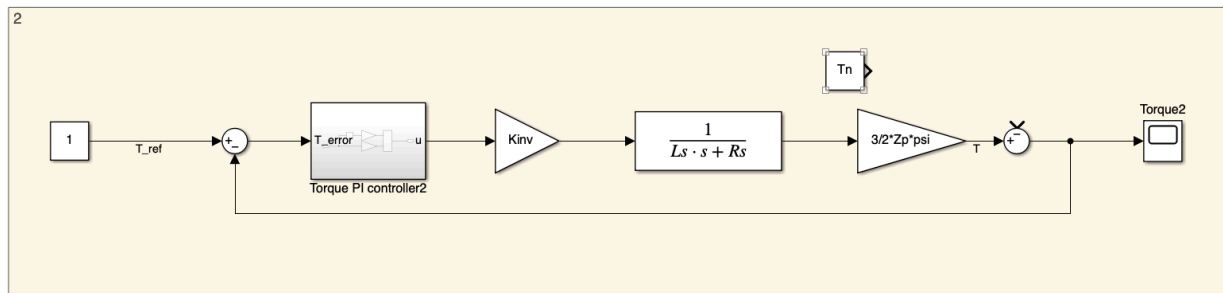


Figure 4 — Simulink Model

Obtain transient responses $M(t)$ to a step torque reference.

1.2.2 Simulation Result

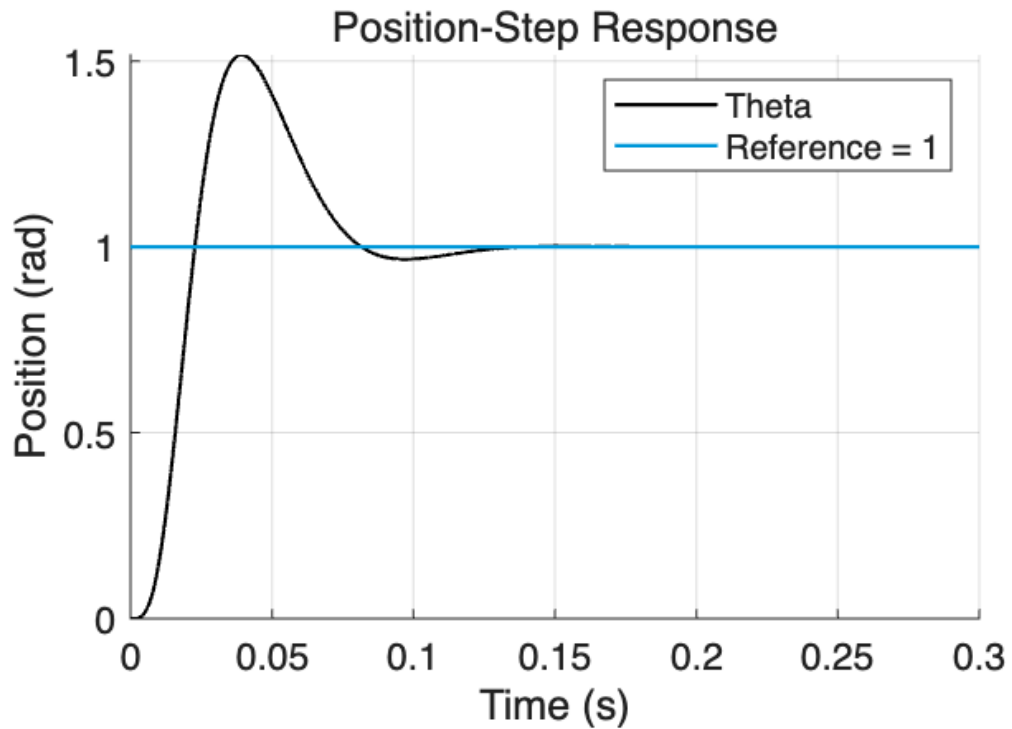


Figure 5 — Step response

1.2.3 Speed Loop

Tune the controller using Magnitude Optimum (MO) method.

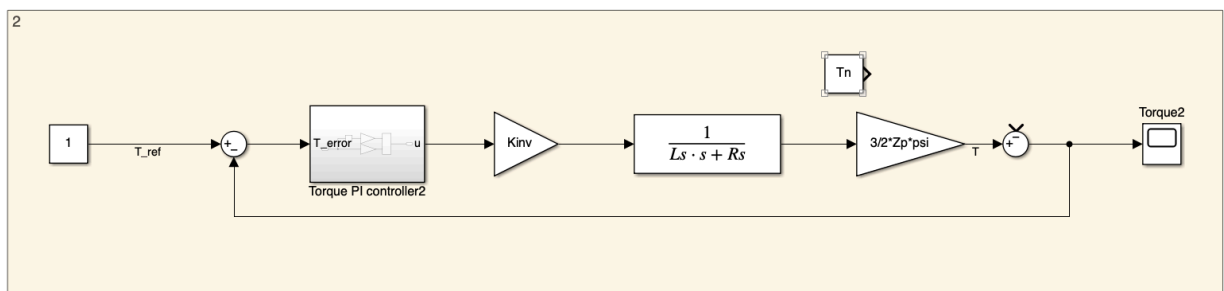


Figure 6 — Simulink Model

Obtain and plot the speed response $\omega(t)$ for a step reference.

1.2.4 Simulation Result

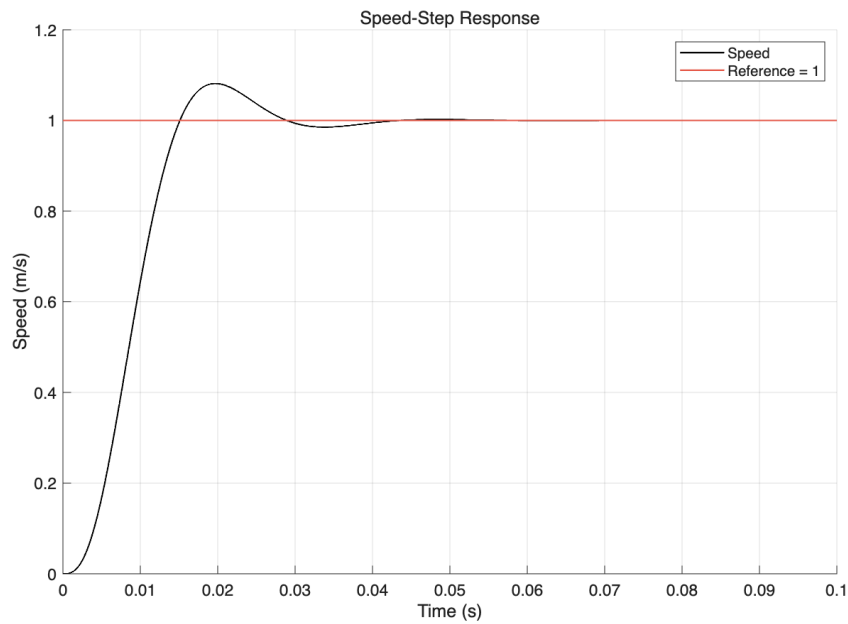


Figure 7 — Step response

1.2.5 Position Loop

Tune the controller using Symmetric Optimum (SO) method.

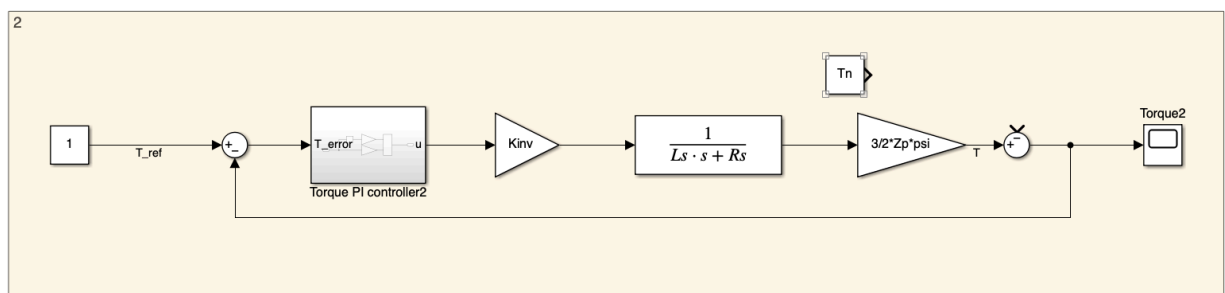


Figure 8 — Simulink Model

Obtain the position response $\theta(t)$ for a step command.

1.2.6 Simulation Result

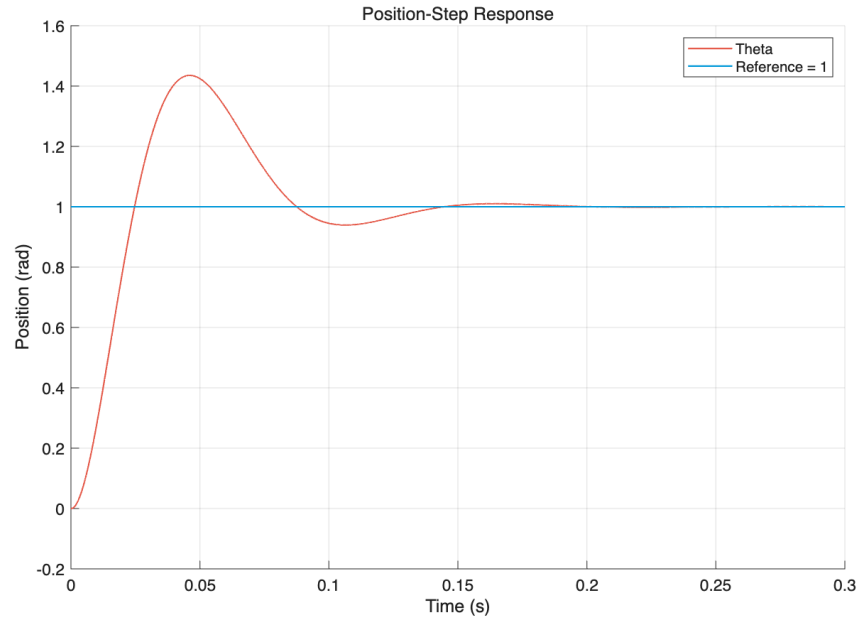


Figure 9 — Step response

1.3 Task 3 : Vector Control (Field-Oriented Control) of SM

1.3.1 Simulink Models

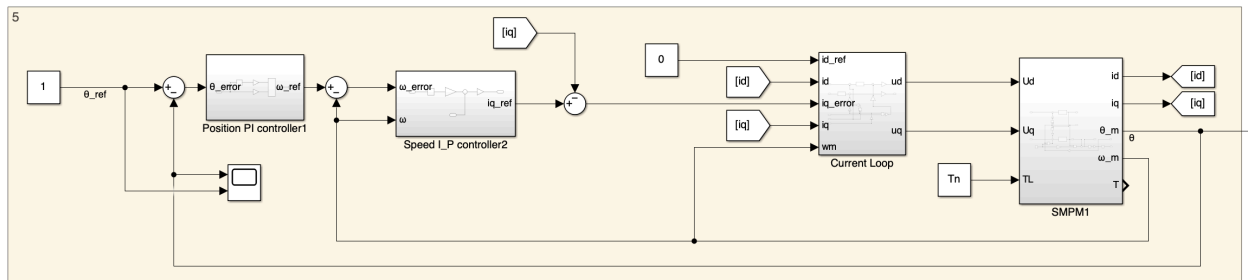


Figure 10 — Field-Oriented Control

1.3.2 Simulation Results

Compare results with the cascaded system and draw conclusions about dynamic performance and accuracy.

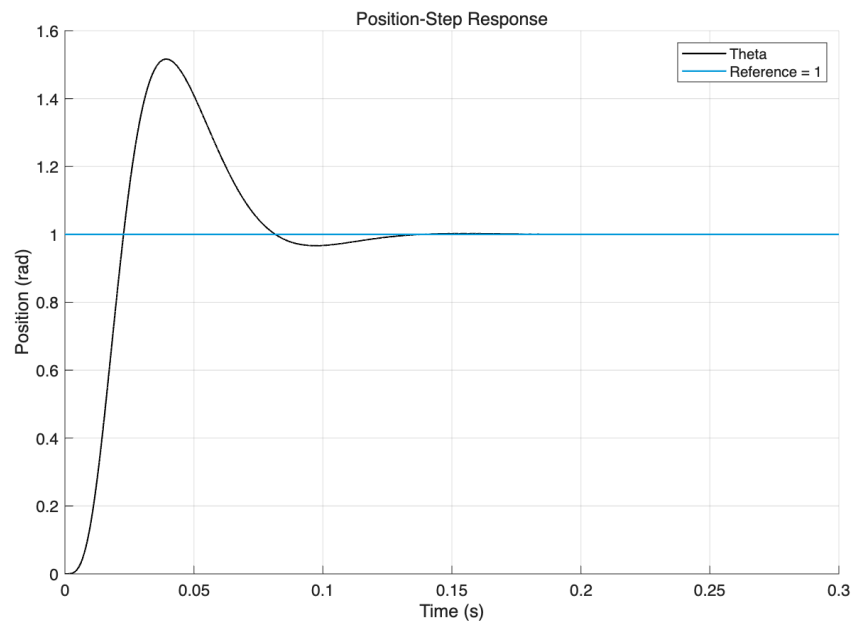


Figure 11 — Step response

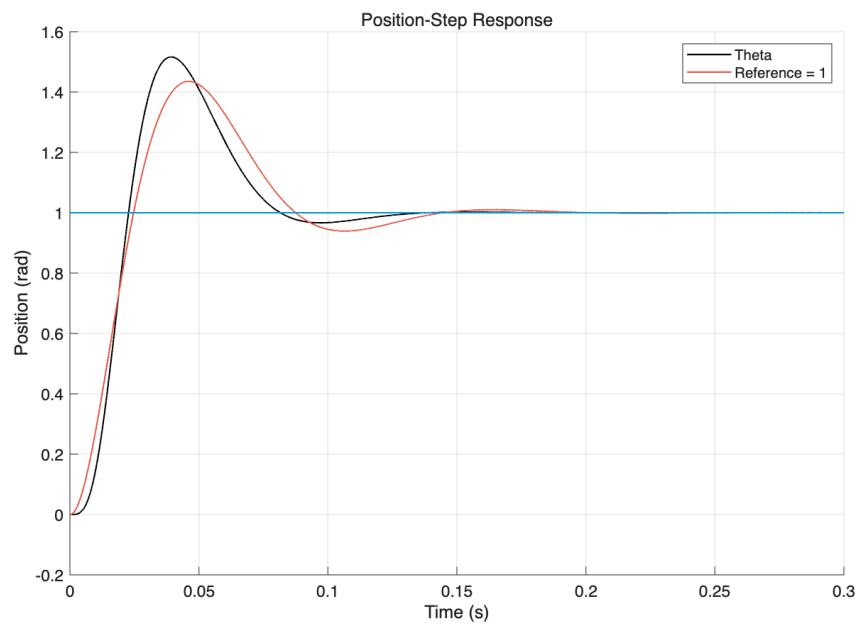


Figure 12 — Comparison of Position Responses

2 CONCLUSION

Based on the comprehensive laboratory work conducted, it is concluded that both cascaded control and vector control strategies were successfully designed and implemented for the synchronous motor drive system. The cascaded control structure provided a straightforward and effective method for regulating torque, speed, and position, demonstrating satisfactory performance with well-tuned PI and P controllers using the BP, MO, and SO methods. However, the vector control approach, with its field-oriented principle and independent control of d-axis and q-axis currents, proved superior in dynamic performance, offering faster response, better decoupling, and enhanced accuracy in torque production. The comparison highlights that while cascaded control is simpler to implement, vector control is more suitable for high-performance applications requiring precise and robust operation across a wide speed range. The practical insights gained from analyzing the transient responses and control quality underscore the importance of selecting the appropriate control strategy based on specific application requirements.